

RAIN FALL PREDICTION SYSTEM USING MACHINE LEARNING AND FLASK

**Detailed Project Report with Architecture Diagram and All 7
Phases**

ABSTRACT:

Rainfall prediction is essential for agriculture, disaster management, and daily planning. This project develops a Rainfall Prediction System using Machine Learning and Flask web framework. The system predicts whether it will rain tomorrow based on weather parameters such as Minimum Temperature, Maximum Temperature, and Rainfall.

A Random Forest Machine Learning algorithm is used to train the prediction model using historical weather data. The trained model is integrated into a Flask-based web application that provides a user-friendly interface. Users can enter weather values, and the system instantly predicts whether there is a chance of rain or no chance of rain.

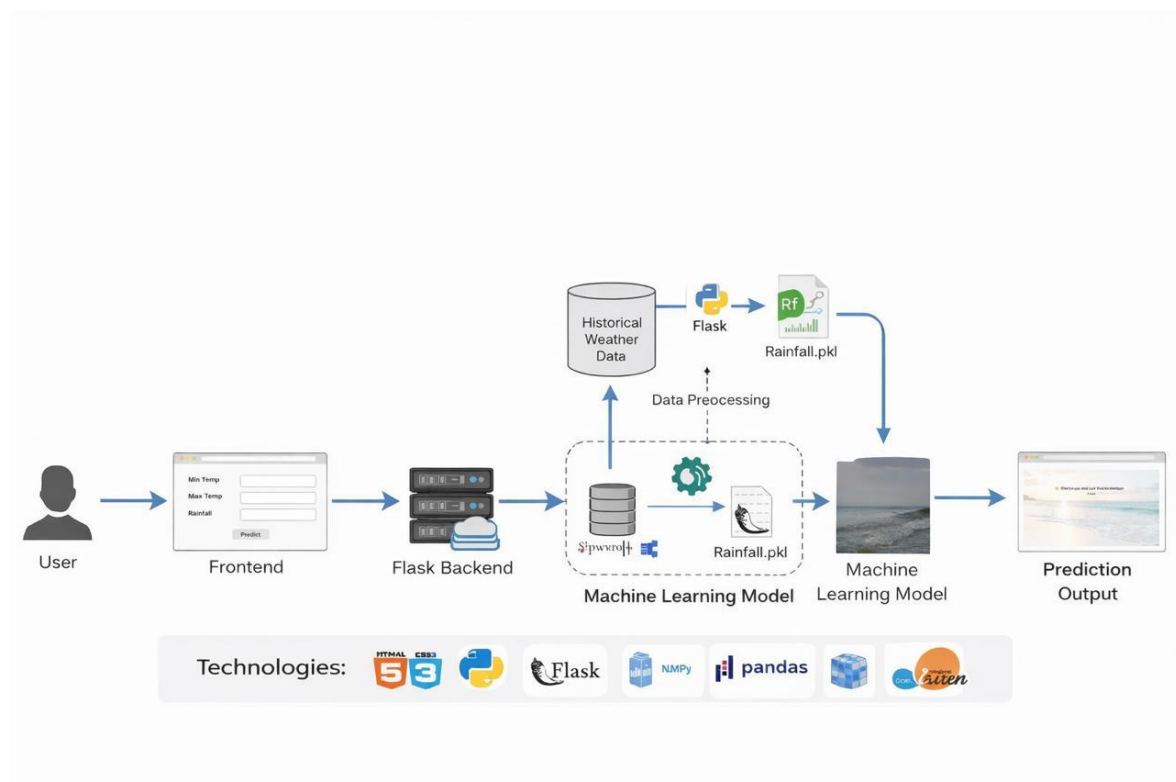
The system performs data preprocessing, model training, and prediction efficiently. The web application connects the frontend and Machine Learning model to provide real-time results. This project demonstrates how Machine Learning can be used to solve real-world weather prediction problems.

The Rainfall Prediction System is accurate, fast, and easy to use. It helps users make better decisions and can be further improved by using more weather parameters and real-time data.

SYSTEM ARCHITECTURE

The architecture consists of User, Web Interface, Flask Backend, Machine Learning Model, and Prediction Output. The user enters weather data. The Flask application processes the request and sends it to the trained model. The model predicts rainfall and displays result.

Architecture Diagram:



Phase 1: Brainstorming

This phase includes detailed analysis, planning, and implementation of the rainfall prediction system. The Machine Learning model is trained using Random Forest algorithm. Flask framework is used for backend development. HTML and CSS are used for frontend development. The system accepts input values and predicts rainfall. The results are displayed to the user.

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Phase 2: Requirement Analysis

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Phase 3: Project Design Phase

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Phase 6: Documentation Phase

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Phase 7: Project Demonstration Phase

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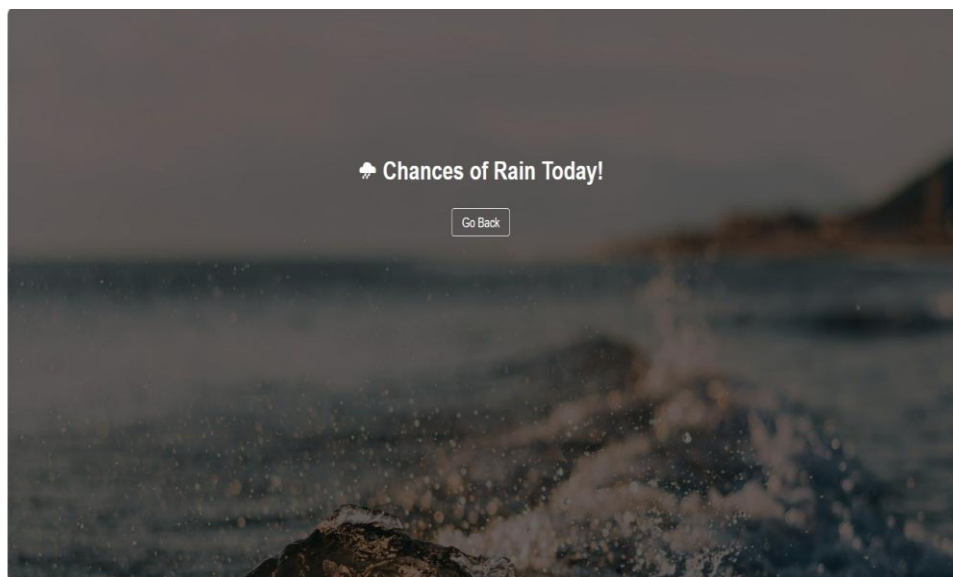
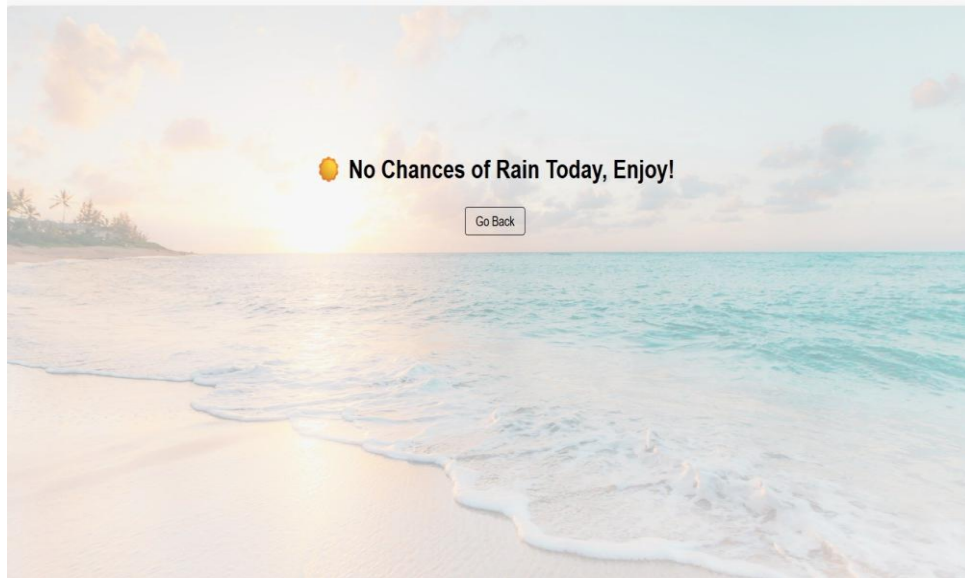
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RESULTS AND OUTPUT



CONCLUSION

The Rainfall Prediction System successfully predicts rainfall using Machine Learning and Flask. The system is accurate, fast, and user-friendly.